

Zigbee Task Sheet

Task 1:

- Payload RAM (127×8 bits) & Input Mux
- Zero-Padding & Payload Framer
- Demux (I/Q bit splitter)
- Symbol Mappers (ROM-based Walsh–Hadamard encoders for I/Q paths)
- Interleaver (250 kbps path)
- **Unit Verification (Verify Task 3):** Write self-checking unit testbenches for the CSK Generator and chirp ROM.

Task 2:

- Preamble / SFD Generator ROM
- Top-Level Controller State Machine (FSM) to sequence Preamble → SFD → PHR → Payload → Done
- QPSK Mapper
- DQPSK Differential Encoder (4-symbol delay with 4 flip-flops + complex multiplier)
- **Unit Verification (Verify Task 1):** Write unit testbenches and generate test vectors for the Symbol Mapper and Interleaver.

Task 3:

- Analyze MATLAB floating-point vs. fixed-point models.
- Justify internal word lengths using the Mean Square Error (MSE) target ($\$MSE < 0.005\$$).
- CSS Symbol / Chirp Sequence ROM (storing the 4 chirp sequences / raised-cosine parameters).
- CSK Generator & Complex Multiplier Modulator.
- **Unit Verification (Verify Task 2):** Write unit testbenches for the DQPSK encoder and verify controller state transitions against MATLAB timing specifications.

Task 4: System Verification

- **Testbench:** Write system test bench for multiple payload lengths (including edge cases like length 0, 1, and 127).
- **MATLAB Verification:** Calculate the overall MSE between the top-level RTL output stream (Tx_real, Tx_imag) and the fixed-point MATLAB reference to guarantee correctness.